A Comparative Study of Transformer and Hybrid Models for Fake News Detection

A thesis submitted in partial fulfilment of the requirements for the degree of

Master of Computer Application

by

Md Akash Hossain

Registration No. 015670 of 2019-2020

Roll No. 250652005

Under the supervision and guidance of

Parijata Majumdar

Assistant Professor

Department of Information Technology

Tripura University, Suryamaninagar



Department of Information Technology

Tripura University

Suryamaninagar, Tripura, India - 799022

July, 2026

Dedicated to my parents

Abstract

Fake news detection is the task of employing computational tools such as Natural Language Processing (NLP) and Machine Learning (ML) to identify and categorize false or misleading information presented as genuine news. The rapid proliferation of misinformation through digital and social media channels poses serious threats to public perception, democratic governance, and the reliability of online information, creating a pressing need for automated detection systems capable of handling large volumes of unstructured text at scale.

This thesis proposes a layered fake news detection framework that combines NLP-based text preprocessing, multiple text representation techniques, and a comparative evaluation of classical, sequential, and transformer-based classification models. During preprocessing, noise removal, tokenization, stopword elimination, and lemmatization are applied to standardize raw news text before feature extraction. Text is then represented numerically using techniques ranging from TF-IDF and Bag-of-Words to Word2Vec, GloVe, and contextual BERT embeddings.

Three classification models are systematically compared on multiple benchmark datasets, namely LIAR and ISOT, to assess both raw performance and cross-dataset generalization: a standalone Bidirectional Long Short-Term Memory (BiLSTM) network, a fine-tuned Bidirectional Encoder Representations from Transformers (BERT) model, and a hybrid BERT + BiLSTM architecture in which BERT's contextual token embeddings are fed into a BiLSTM layer to additionally model sequential dependencies. Experimental results show that the standalone BiLSTM model performs poorly, achieving only 50% accuracy and an F1-score of 0.3333, a limitation attributed to a restricted number of training epochs and the absence of extensive hyperparameter tuning. In contrast, the BERT model substantially outperforms the sequential baseline, reaching up to 95% accuracy with a precision of 0.9545 and a recall of 0.9500. The hybrid BERT + BiLSTM model matches BERT's performance almost exactly, indicating that, under the present experimental setup, the additional sequential layer neither improves nor degrades the transformer's discriminative power.

These findings demonstrate the clear efficiency of transformer-based architec-

tures for fake news detection over purely sequential deep learning models, and confirm that the resulting system is scalable enough for practical deployment in scenarios such as social media monitoring and automated fact-checking. The thesis concludes by outlining future directions that integrate Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Explainable AI (XAI) techniques to further improve detection accuracy, evidence grounding, and the interpretability of model predictions for end users such as journalists and fact-checkers.

Keywords: Fake News Detection, Natural Language Processing, Machine Learning, Deep Learning, BERT, BiLSTM, Transformers, Explainable AI, Misinformation.

Declaration

I certify that

- The work contained in the dissertation is original and has been done by myself under the general supervision of my supervisor.
- The work has not been submitted to any other Institute for any degree or diploma.
- I have followed the guidelines provided by Tripura University in writing the thesis.
- I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the university.
- Whenever I have used materials (data, theoretical analysis, and text) from other sources, I have given due credit to them by citing them in the text of the dissertation and giving their details in the references.
- I have not use any AI websites (eg: chat GPT) for paraphrasing sentences.
- I have not plagiarised any content from other sources.

Md Akash Hossain

Department of Information Technology
Tripura University
Suryamaninagar, West Tripura, India, 799022.
Phone: +91 XXXXXXXXXX

Parijata Majumdar
Assistant Professor
E-Mail : parijatamajumdar@tripurauniv.ac.in

Certificate

This is to certify that the thesis entitled **"A Comparative Study of Transformer and Hybrid Models for Fake News Detection"** submitted to Tripura University in the Department of Information Technology in partial fulfillment of the award of the degree of **Master of Computer Application** is a record of research work carried out by **Md Akash Hossain**, Registration Number **015670 of 2019-2020**, **Roll No. 250652005** under my supervision and guidance.

All helps received by him from various sources have been duly acknowledged. No part of this thesis has been submitted elsewhere for award of any other degree.

Signature of Supervisor
(Parijata Majumdar)
Assistant Professor
Department of Information Technology
Tripura University
Suryamaninagar, Tripura, India, 799022

Department of Information Technology
Tripura University
Suryamaninagar, West Tripura, India, 799022.
Phone: +91 9774829599

Dr. Jayanta Pal
Assistant Professor
E-Mail : jayantapal@tripurauniv.ac.in

Certificate

Certified that the thesis entitled **"A Comparative Study of Transformer and Hybrid Models for Fake News Detection"** submitted by **Md Akash Hossain**, Registration Number **015670 of 2019-2020**, **Roll No. 250652005** to Tripura University in the Department of Information Technology is a record of research work carried out under the supervision and guidance of **Parijata Majumdar** is worthy of consideration for the award of the degree of **Master of Computer Application** of the University.

Signature of Project Coordinator
(Dr. Jayanta Pal)
Assistant Professor
Department of Information Technology
Tripura University
Suryamaninagar, Tripura, India, 799022

Department of Information Technology
Tripura University
Suryamaninagar, West Tripura, India, 799022.
Phone: 03712-275356

Prof. Swanirbhar Majumder
Head of the Department
E-Mail : hod_it@tripurauniv.ac.in

Certificate

Certified that the thesis entitled **"A Comparative Study of Transformer and Hybrid Models for Fake News Detection"** submitted by **Md Akash Hossain**, Registration Number **015670 of 2019-2020**, **Roll No. 250652005** to Tripura University in the Department of Information Technology is a record of research work carried out under the supervision and guidance of **Parijata Majumdar** is worthy of consideration for the award of the degree of **Master of Computer Application** of the University.

Signature of HoD
(Prof. Swanirbhar Majumder)
Head of the Department
Department of Information Technology
Tripura University
Suryamaninagar, Tripura, India, 799022



Certificate

This is to certify that the thesis entitled **"A Comparative Study of Transformer and Hybrid Models for Fake News Detection"** submitted by **Md Akash Hossain**, Registration Number **015670 of 2019-2020**, Roll No, **250652005** to Tripura University in the Department of Information Technology in partial fulfillment of the requirements for the award of the degree of **Master of Computer Application** has been examined by us on ..... and found to be satisfactory.

The Committee recommends for award of the degree of Master of Computer Application.

Signature of Internal Examiner Signature of External Examiner

Acknowledgment

It's my great pleasure to express my gratitude to all of them who have provided me guidance and support in bringing out the successful completion of my Master of Computer Application program at Tripura University. It would not have been possible for me to achieve success with my effort alone. There were a lot of people who extended their supportive hands towards me in the way of making my work a success. I would like to thank everyone who supported and assisted me while carrying out this work at Tripura University.

First of all, I would like to thank my supervisor **Parijata Majumdar** for her constant support, trust, valuable feedback, encouragement and innumerable advice. She gave freedom to pursue my ideas and work at my own pace and was always available to discuss various problems on the way. Her encouragement and guidance have provided a good basis for completion of my research work.

I am highly obliged to all the faculty members and non-teaching staff of the Department of Information Technology for their support and encouragement. I also thank **Prof. Ganga Prasad Prasain**, Vice Chancellor of Tripura University and **Prof. Swanirbhar Majumder**, Head, Department of Information Technology for providing excellent computing and other facilities without which this work could not achieve its quality goal.

I will ever remain grateful to all my family members for their unconditional love and support. I am deeply indebted to them who has been constantly encouraging and supporting me in every walk of my life.

Finally, I would like to thank all those who have directly or indirectly helped me in different capacities to complete my research work.

Md Akash Hossain

Contents

1 Introduction **1**

1.1 Challenges and Issues 2

1.2 Motivation of the Research 3

1.3 Research Objectives 3

1.4 Thesis Contributions 4

1.5 Thesis Organization 5

2 Background Details **7**

2.1 Introduction 7

2.2 Fake News and the Role of Machine Learning 7

2.3 NLP Preprocessing Pipeline 8

2.4 Text Representation Techniques 9

2.5 Classical Machine Learning Classifiers 10

2.6 Deep Learning Sequence Models: LSTM and BiLSTM 10

2.7 Transformer Architecture and BERT 11

2.8 Summary 12

3 Related Work **13**

3.1 Introduction 13

3.2 Classical Machine Learning Approaches 22

3.3 Deep Learning and Hybrid Architectures 23

3.4 Benchmark Datasets, Surveys, and Empirical Studies 23

3.5 Large Language Models and Explainability 24

3.6 Summary 24

4 Proposed Approach **26**

4.1 Introduction 26

4.2 Problem Statement 26

4.3 System Model 27

4.4 Proposed BERT + BiLSTM Detection Framework 28

4.4.1 Proposed Algorithm 29

4.4.2 Design Rationale: Why Combine BERT and BiLSTM? . . . 30

4.5 Summary 31

5 Experimental Results **32**

5.1 Introduction 32

5.2 Simulation Setup 32

5.2.1 Evaluation Metrics 32

5.2.2 Dataset Description 33

5.2.3 Hyperparameter Configuration 34

5.3 Simulation Output 34

5.4 Result Discussion 36

5.5 Summary 38

6 Conclusion and Future Direction **39**

6.1 Conclusion 39

6.2 Future Research Direction 40

List of Figures

4-1 End-to-end fake news detection system pipeline, from user input through data collection, NLP preprocessing, feature extraction, classification, and output reporting 27

4-2 Detailed architecture of the proposed BERT + BiLSTM fake news detection model 29

5-1 Confusion Matrix for BERT, BiLSTM, and BERT+BiLSTM models across all datasets 35

5-2 ROC Curves for BERT, BiLSTM, and BERT+BiLSTM models across all datasets 36

5-3 Precision–Recall Curves for BERT, BiLSTM, and BERT+BiLSTM models across all datasets 36

5-4 Loss Curves for BERT, BiLSTM, and BERT+BiLSTM models across all datasets 37

List of Tables

3.1 Comparative Literature Review of Fake News Detection Methods . 14

5.1 Dataset Description 33

5.2 Hyperparameter Configuration of Models 34

5.3 Performance Comparison of Fake News Detection Models 34

5.4 Performance Comparison Across Multiple Datasets 35

List of Algorithms

1 Fake News Detection using the Proposed BERT + BiLSTM Model . 30

Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
LLM	Large Language Model
XAI	Explainable Artificial Intelligence
RAG	Retrieval-Augmented Generation
BERT	Bidirectional Encoder Representations from Transformers
LSTM	Long Short-Term Memory
BiLSTM	Bidirectional Long Short-Term Memory
RNN	Recurrent Neural Network
CNN	Convolutional Neural Network
GNN	Graph Neural Network
SVM	Support Vector Machine
LR	Logistic Regression
RF	Random Forest
TF-IDF	Term Frequency-Inverse Document Frequency
BoW	Bag-of-Words
GloVe	Global Vectors for Word Representation
CLS	Classification Token
SEP	Separator Token
ReLU	Rectified Linear Unit
Adam	Adaptive Moment Estimation
AdamW	Adam with Decoupled Weight Decay
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative

Abbreviation	**Full Form**
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
PR	Precision–Recall
LIME	Local Interpretable Model-agnostic Explanations
SHAP	SHapley Additive exPlanations
LIAR	LIAR Benchmark Dataset (PolitiFact short statements)
ISOT	ISOT Fake News Dataset (University of Victoria)
API	Application Programming Interface
GPU	Graphics Processing Unit
QA	Question Answering
IR	Information Retrieval

Symbols and Notations

T	Raw input news text
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
$E_{[CLS]}$	BERT embedding of the classification token
$E_{[SEP]}$	BERT embedding of the separator token
AUC	Area under the ROC curve

Chapter 1

Introduction

Fake news detection is the automated process of identifying and categorizing false, misleading, or fabricated information that is deliberately or unintentionally portrayed as genuine news, using computational methods drawn from Natural Language Processing (NLP) and Machine Learning (ML) [15, 16]. The emergence and rapid growth of digital media platforms has led to an unprecedented increase in the volume, velocity, and reach of information generated and shared online every day. News, opinion, and rumour now travel through the same channels, often indistinguishable from one another at first glance, and readers are left to judge credibility with little more than a headline and a few seconds of attention.

This state of affairs creates serious challenges, including the manipulation of public perception, interference with democratic processes such as elections, and a gradual erosion of confidence in previously trusted sources of information [15, 16]. Empirical studies have shown that false information spreads significantly faster and reaches a wider audience on social media than verified, factual news [18]. Given this pace, it becomes practically infeasible for human journalists and professional fact-checkers to manually verify every claim before it spreads beyond control.

The consequences of unchecked misinformation extend well past individual misunderstanding; large-scale disinformation campaigns have been linked to social unrest, financial market manipulation, and public health harms. Consequently, building automated systems that can detect fake news reliably, at scale, and in near real time has become an important and urgent research problem, motivating the work undertaken in this thesis.

1.1 Challenges and Issues

The term "fake news" is not a single, homogeneous category of content. It is an umbrella term that spans multiple distinct classes of information, each with its own motive, structure, and linguistic signature. Broadly, the literature identifies five recurring types of fake news:

- **False News Stories:** Entirely fabricated information deliberately designed to deceive readers.
- **Manipulated News:** Genuine information that has been selectively edited, distorted, or taken out of context.
- **Satirical News:** Exaggerated or humorous content that is not intended to deceive but may be mistaken for factual reporting.
- **Clickbait:** News stories with sensational or misleading headlines that do not match their actual content.
- **Propaganda:** Biased or slanted information deliberately produced to advance a particular agenda or cause.

This diversity is precisely what makes automatic detection difficult. A model trained to recognize the linguistic markers of one category, such as sensational clickbait headlines, may fail entirely when confronted with subtly manipulated but factually-grounded reporting, or with satire that mimics a legitimate news outlet's tone. Detecting fake news therefore requires models that can capture deep contextual and semantic relationships in text, rather than relying on superficial keyword or stylistic cues alone.

A second layer of difficulty comes from the modelling side. Classical machine learning algorithms such as Logistic Regression, Support Vector Machines (SVM), Naive Bayes, and Random Forest have historically been applied to this task using hand-engineered statistical features (e.g., TF-IDF vectors). These models are simple, interpretable, and computationally cheap, but they cannot capture word order or long-range semantic dependency, which limits their ceiling on accuracy. Sequential deep learning models such as Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM) networks were introduced to address this gap by explicitly modelling sequences of words, yet their effectiveness can be undermined when contextual relationships are complex, when training data is limited,

or when training time and hyperparameter tuning are restricted – a limitation that is directly observed in the experimental results of this thesis (Chapter 5).

Transformer-based models, most notably BERT (Bidirectional Encoder Representations from Transformers), have been shown to outperform these earlier sequential approaches by producing deeply bidirectional, contextualized representations of text in a single forward pass [5, 8]. Where sequential models process text step by step, BERT attends to the entire input context simultaneously through self-attention, giving it a considerably richer view of semantic relationships between words. This contrast between sequential and attention-based contextual modelling is one of the central technical challenges this thesis investigates.

1.2 Motivation of the Research

The motivation for this research is twofold. First, from a societal standpoint, the scale and speed at which misinformation now spreads make manual fact-checking an inadequate defence on its own; an automated, scalable first line of detection is necessary to support (not replace) human fact-checkers and platform moderators. Second, from a technical standpoint, the field has moved rapidly from classical statistical text representations (Bag-of-Words, TF-IDF) through static embeddings (Word2Vec, GloVe) to deep contextual embeddings (BERT), and it is not always clear, for a given dataset and task, whether the additional complexity of transformer and hybrid architectures is actually justified by a corresponding improvement in accuracy.

This thesis is therefore motivated by the need to systematically and empirically compare a purely sequential deep learning model (BiLSTM), a purely transformer-based model (BERT), and a hybrid model that combines both (BERT + BiLSTM), across multiple benchmark datasets, so that the trade-offs between model complexity, training cost, and detection performance can be clearly understood. Such a comparison directly informs practical deployment decisions for real-world fake news detection systems, including social media monitoring pipelines and automated fact-checking assistants.

1.3 Research Objectives

The specific objectives of this research are as follows:

1. To design a multi-stage fake news detection pipeline that integrates NLP-based text preprocessing, multiple feature extraction/representation techniques, and a classification layer.

2. To implement and train three classification models – a standalone BiLSTM network, a fine-tuned BERT model, and a hybrid BERT + BiLSTM model – under comparable training conditions.

3. To evaluate and compare the three models using standard classification metrics (accuracy, precision, recall, and F1-score) as well as visual diagnostics (confusion matrices, ROC curves, precision–recall curves, and loss curves).

4. To assess the generalization ability of the proposed models across multiple benchmark datasets, namely LIAR and ISOT.

5. To identify the practical and architectural trade-offs between sequential, transformer-based, and hybrid models for fake news detection, and to outline directions for future improvement using large language models and explainable AI.

1.4 Thesis Contributions

This thesis makes the following key contributions:

- Proposes a layered architecture for fake news detection that combines NLP preprocessing, multiple feature extraction techniques, and a comparative classification framework.

- Presents a systematic experimental comparison of BiLSTM, BERT, and a hybrid BERT + BiLSTM model across multiple benchmark datasets.

- Demonstrates, with quantitative evidence, the superiority of transformer-based models over purely sequential deep learning models for this task under constrained training budgets.

- Investigates and documents the specific limitations of sequential models and their hybrid variants when trained with a small number of epochs and without extensive hyperparameter tuning.

- Identifies promising directions for future research, including the integration of Large Language Models (LLMs), Retrieval-Augmented Generation

(RAG), and Explainable AI (XAI) techniques into fake news detection pipelines.

1.5 Thesis Organization

The remainder of this thesis is organized into six chapters, structured as follows.

Chapter 1: Introduction

This chapter has introduced the problem of fake news detection, its societal significance, the challenges inherent in automatically distinguishing genuine from fabricated content, the motivation behind this research, and the specific objectives and contributions of the thesis.

Chapter 2: Background Details

This chapter presents the foundational background required to understand the rest of the thesis, including an overview of the fake news landscape, the standard NLP text preprocessing pipeline, classical machine learning classifiers, sequential deep learning models (LSTM and BiLSTM), transformer architectures and BERT, and the text representation techniques (TF-IDF, Bag-of-Words, Word2Vec, GloVe, and contextual embeddings) used throughout this work.

Chapter 3: Literature Survey

This chapter provides an extensive review of existing literature on fake news detection, organized thematically from classical machine learning techniques, through deep learning and hybrid architectures, to transformer-based and evidence-aware methods, benchmark dataset papers, and recent work on large language models and explainability.

Chapter 4: Proposed Approach

This chapter describes the proposed multi-stage fake news detection system in detail, including the problem statement, the overall system model and pipeline,

the detailed BERT + BiLSTM model architecture, the underlying algorithm, and the rationale for combining a transformer encoder with a sequential BiLSTM layer.

Chapter 5: Experimental Results

This chapter presents the experimental setup, datasets, evaluation metrics, and hyperparameter configuration used to train and evaluate the three models, followed by a detailed presentation and discussion of the comparative results, confusion matrices, ROC curves, precision–recall curves, and training/validation loss curves.

Chapter 6: Conclusion and Future Direction

This chapter summarizes the key findings of the thesis, discusses their implications for practical fake news detection systems, and outlines future research directions involving large language models, Retrieval-Augmented Generation, Explainable AI, multilingual and multi-modal detection, and federated deployment.

Chapter 2

Background Details

2.1 Introduction

This chapter presents the technical background required to follow the remainder of the thesis. It begins with an overview of the fake news landscape and the role machine learning and deep learning play in addressing it, then walks through the standard NLP preprocessing pipeline, the major text representation techniques used to convert raw text into numeric form, the classical machine learning classifiers historically applied to this task, the sequential deep learning models (LSTM and BiLSTM) that improved on them, and finally the transformer architecture and BERT, which form the core of the model proposed in Chapter 4.

2.2 Fake News and the Role of Machine Learning

Fake news detection sits at the intersection of NLP, ML, and social computing. At its core, the task is a text classification problem: given a piece of text (a headline, article body, or social media post), predict whether it represents genuine or fabricated/misleading news. What makes the problem hard in practice is that this classification decision often requires world knowledge, an understanding of context, source credibility, and sometimes multi-modal evidence (images, videos, metadata) that a text-only classifier does not have direct access to.

Machine learning and deep learning techniques have proven effective for

this task because they can automatically learn discriminative patterns from large labelled datasets rather than relying on manually crafted rules. Classical ML algorithms such as Logistic Regression, Support Vector Machines (SVM), Naive Bayes, and Random Forest operate on hand-engineered statistical features extracted from text (most commonly TF-IDF vectors) and were the dominant approach before deep learning became practical for NLP at scale. Deep learning models, by contrast, learn their own feature representations directly from data, which allows them to capture more nuanced linguistic patterns, at the cost of requiring more data, compute, and careful tuning.

2.3 NLP Preprocessing Pipeline

Raw news text is noisy: it contains inconsistent casing, punctuation, HTML artifacts, and words that carry little discriminative signal. Before any feature extraction or model training can take place, the text must be cleaned and normalized. The standard NLP preprocessing pipeline used in this thesis consists of the following steps:

- **Noise Removal:** Stripping HTML tags, special characters, URLs, and other non-textual artifacts from the raw input.
- **Lowercasing:** Converting all text to a consistent case so that, e.g., "News" and "news" are treated identically.
- **Tokenization:** Splitting text into individual tokens (words or sub-word units), which form the atomic units consumed by downstream models.
- **Stopword Removal:** Removing high-frequency, low-information words (e.g., "the", "is", "and") that do not contribute to distinguishing fake from real news.
- **Stemming and Lemmatization:** Reducing words to their root or base form (e.g., "running" $\rightarrow$ "run"), which reduces vocabulary size and helps models generalize across word forms.
- **Padding / Truncation:** Normalizing the length of tokenized sequences to a fixed maximum length so that they can be processed in fixed-size batches by neural network models.

For transformer models such as BERT, tokenization is performed using a model-specific sub-word tokenizer (WordPiece), and two special tokens are introduced: a classification token, `[CLS]`, prepended to every sequence and used to aggregate sequence-level information, and a separator token, `[SEP]`, used to mark sequence boundaries. Each tokenized sequence is accompanied by an attention mask that indicates which positions correspond to real tokens (1) versus padding (0), so that the model does not attend to padded positions.

2.4 Text Representation Techniques

Once text has been cleaned and tokenized, it must be converted into a numeric representation before it can be consumed by a machine learning model. This thesis makes use of several representation techniques, of increasing representational richness:

- **Bag-of-Words (BoW):** Represents a document as an unordered multiset of its word counts, discarding grammar and word order entirely.
- **TF-IDF (Term Frequency-Inverse Document Frequency):** Weights word counts by how informative a word is across the corpus, down-weighting common words and up-weighting rare, discriminative ones.
- **Word2Vec:** Learns dense, fixed-length vector representations of words such that words appearing in similar contexts have similar vectors, capturing some notion of semantic similarity.
- **GloVe (Global Vectors for Word Representation):** Learns word embeddings from global word-word co-occurrence statistics across a corpus, producing embeddings similar in spirit to Word2Vec but derived from a different (count-based) objective.
- **BERT Contextual Embeddings:** Unlike the previous techniques, which assign a single fixed vector to each word regardless of context, BERT produces a different embedding for a word depending on its surrounding sentence, capturing polysemy and richer contextual meaning.

The first three (BoW, TF-IDF) are purely statistical and are typically paired with classical ML classifiers, while Word2Vec and GloVe provide dense embeddings suitable as input to sequential deep learning models such as

LSTM/BiLSTM. BERT embeddings, being contextual, are used directly with transformer-based classifiers and, in the hybrid model studied in this thesis, are additionally passed into a BiLSTM layer.

2.5 Classical Machine Learning Classifiers

Before deep learning became the dominant paradigm for text classification, several classical supervised learning algorithms were widely applied to fake news detection using statistical text features:

- **Logistic Regression:** A linear model that estimates the probability of a binary outcome (real vs. fake) as a function of the input features; simple, fast, and interpretable.
- **Support Vector Machine (SVM):** Finds a maximum-margin hyperplane that separates the two classes in feature space, and performs well on high-dimensional sparse text features such as TF-IDF vectors.
- **Naive Bayes:** A probabilistic classifier based on Bayes' theorem with a (naive) assumption of conditional independence between features, historically popular for text classification due to its speed and reasonable performance with sparse features.
- **Random Forest:** An ensemble of decision trees trained on bootstrapped samples of the data, which tends to be more robust to noise and overfitting than a single decision tree.

These models are computationally cheap and easy to interpret, but they treat text as an unordered collection of statistical features and cannot capture word order or long-range semantic dependencies, which fundamentally limits their ceiling of performance on nuanced classification tasks like fake news detection.

2.6 Deep Learning Sequence Models: LSTM and BiLSTM

Long Short-Term Memory (LSTM) networks are a class of Recurrent Neural Networks (RNNs) specifically designed to model sequential data while mitigating the

vanishing gradient problem that plagues vanilla RNNs over long sequences. An LSTM cell maintains an internal memory state that is selectively updated, forgotten, and output at each time step through a set of learned gates (input, forget, and output gates), allowing the network to retain relevant information over longer spans of text than a standard RNN.

A standard LSTM processes a sequence in a single direction, typically left to right, meaning that at any given position, the model has only seen the preceding context. A Bidirectional LSTM (BiLSTM) addresses this limitation by running two LSTM layers over the input sequence in parallel, one processing it left-to-right and the other right-to-left, and concatenating (or otherwise combining) their hidden states at each position. This gives the BiLSTM access to both past and future context, which is particularly useful for understanding a word's role in a sentence.

Despite this improvement, BiLSTM networks still process text strictly sequentially at inference time, which limits parallelization during training, and their capacity to model very long-range dependencies remains weaker than attention-based architectures. Furthermore, as observed experimentally in Chapter 5, BiLSTM models trained with a limited number of epochs and without extensive hyperparameter tuning can converge poorly, resulting in substantially lower accuracy than transformer-based alternatives.

2.7 Transformer Architecture and BERT

The Transformer architecture dispenses with recurrence entirely and instead relies on a self-attention mechanism, which allows every token in a sequence to directly attend to every other token in a single layer, regardless of their distance in the sequence. This removes the sequential bottleneck of RNN-based models and enables far greater parallelization during training, while also making it easier to model long-range dependencies.

BERT (Bidirectional Encoder Representations from Transformers) is built from a stack of Transformer encoder layers (12 layers in the BERT-Base configuration) and is pre-trained on large unlabelled text corpora using two self-supervised objectives: masked language modelling (predicting randomly masked tokens from their context) and next sentence prediction. This pre-training allows BERT to learn deep, bidirectional, general-purpose language representations before it is

ever exposed to a downstream task.

For a downstream classification task such as fake news detection, a pre-trained BERT model is *fine-tuned*: the input news text is tokenized and fed through the pre-trained encoder, producing a contextual embedding for every token, including the special `[CLS]` token, which is conventionally used as an aggregate representation of the entire sequence. A lightweight classification head (typically a dropout layer followed by a fully connected layer with a softmax or sigmoid activation) is then trained on top of this representation to predict the real/fake label, while the underlying BERT weights are also updated (fine-tuned) using a comparatively small learning rate, since they are already well-initialized from pre-training.

In the hybrid architecture examined in this thesis (Chapter 4), rather than relying solely on the `[CLS]` token, the entire sequence of contextual token embeddings produced by BERT is passed into a BiLSTM layer, allowing the network to additionally model sequential dependencies over BERT's already contextualized representations before final classification.

2.8 Summary

This chapter has covered the technical foundation needed to understand the rest of the thesis: the standard NLP preprocessing pipeline, the range of text representation techniques from simple statistical counts to deep contextual embeddings, classical machine learning classifiers, the sequential LSTM/BiLSTM family of deep learning models, and the Transformer/BERT architecture that underlies the transformer-based models compared in this work. Chapter 3 builds on this foundation with a detailed review of related literature, and Chapter 4 describes how these building blocks are combined into the proposed fake news detection system.

Chapter 3

Related Work

3.1 Introduction

This chapter reviews nineteen peer-reviewed articles that form the basis of this research. The selected articles have been organized thematically, starting with traditional machine learning techniques, moving through deep learning and hybrid architectures, evidence-aware and attention-based methods, benchmark dataset and survey papers, and finally the role of large language models and explainability. All the reviewed articles are summarized in Table 3.1, followed by a thematic discussion.

Table 3.1: Comparative Literature Review of Fake News Detection Methods

Contribution	**Method**	**Advantage**	**Disadvantage**	**Future Scope**
Introduced FakeBERT, a BERT-based deep learning model specifically designed for fake news detection on social media platforms with parallel CNN layers integrated into BERT architecture [5].	BERT + Parallel CNN layers for multi-scale feature extraction on LIAR and ISOT datasets [5].	Achieved 98.90% accuracy; effectively captures contextual semantic relationships in text; outperforms traditional ML models significantly [5].	Computationally expensive; requires large GPU memory; limited to English language content only [5].	Extend to multilingual detection; reduce computational cost using model distillation techniques [5].
Systematically compared multiple classical ML classifiers for fake news detection using linguistic and statistical text features extracted via TF-IDF [1].	Logistic Regression, SVM, Naive Bayes, Random Forest with TF-IDF feature vectors on benchmark news datasets [1].	Simple implementation; low computational cost; interpretable results; suitable for resource-constrained environments [1].	Lower accuracy compared to deep learning; cannot capture semantic context or word order in text [1].	Combine classical ML with deep learning features; incorporate metadata and social context signals [1].

Continued on next page...

Contribution	Method	Advantage	Disadvantage	Future Scope
Proposed a hybrid deep learning architecture combining CNN for local feature extraction and RNN for sequential pattern learning in news text [9].	Hybrid CNN-RNN architecture with GloVe word embeddings trained on ISOT and FAKES datasets [9].	Captures both local and sequential text features simultaneously; better generalization than single-model approaches [9].	Requires significant training data; slow inference time; struggles with short text like tweets and headlines [9].	Apply to real-time social media streams; incorporate attention mechanisms for better interpretability [9].
Analyzed stylometric and linguistic style features that distinguish fake news from credible news using writing patterns and rhetorical structures [11].	Stylometric analysis with SVM classifier; features include sentence length, punctuation patterns, readability scores on news datasets [11].	Does not require external knowledge base; works purely on writing style; lightweight and fast inference [11].	Style-based features are easily mimicked by sophisticated fake news creators; low robustness against adversarial content [11].	Combine style features with semantic embeddings for more robust multi-feature detection systems [11].
Proposed evidence-aware fake news detection by retrieving external web evidence and combining it with claim credibility assessment using attention networks [10].	Attention-based LSTM with external web evidence retrieval and source credibility scoring on Snopes and PolitiFact datasets [10].	Incorporate external knowledge for verification; attention mechanism provides interpretability; better performance on political claims [10].	Dependent on external evidence availability; slow due to web retrieval; fails when evidence is unavailable or biased [10].	Integrate real-time knowledge graphs; improve evidence retrieval with more reliable fact-checking APIs [10].

Continued on next page...

Contribution	Method	Advantage	Disadvantage	Future Scope
Developed a comprehensive fake news benchmark dataset incorporating news content, user social engagement, and spatiotemporal propagation features [16].	Multi-modal dataset construction combining news text, social network graphs, user profiles, and publisher credibility on PolitiFact and GossipCop [16].	Most comprehensive publicly available dataset; enables multi-modal detection research; includes social context signals [16].	Dataset collection is time-consuming; social context features require platform API access which is increasingly restricted [16].	Expand dataset to include video and image modalities; add multilingual fake news samples [16].
Surveyed state-of-the-art deep learning techniques for fake news detection covering transformer models, graph neural networks, and multi-modal approaches [8].	Systematic review of BERT, GPT, GNN, and multi-modal models across multiple benchmark datasets from 2017 to 2022 [8].	Comprehensive coverage of recent methods; identifies research gaps; provides clear taxonomy of detection approaches [8].	Survey only; no novel model proposed; coverage limited to English language studies [8].	Develop unified evaluation benchmark; explore cross-lingual and low-resource fake news detection [8].
Combined BERT contextual embeddings with enhanced LSTM architecture for improved sequential and contextual understanding of fake news text [12].	BERT + enhanced LSTM hybrid model with attention mechanism trained on LIAR and ISOT datasets [12].	Superior performance over standalone BERT or LSTM; attention layer improves interpretability; robust to text length variations [12].	High model complexity; longer training time; requires substantial labeled data for effective fine-tuning [12].	Apply transfer learning to low-resource languages; integrate image features for multi-modal detection [12].

Continued on next page...

Contribution	Method	Advantage	Disadvantage	Future Scope
Investigated psychological and cognitive bias patterns embedded in fake news content using computational linguistics and psycholinguistic feature analysis [2].	Psycholinguistic feature extraction using LIWC tool with ML classifiers; analysis of cognitive bias markers in fake vs. real news articles [2].	Novel perspective combining psychology and NLP; interpretable features; applicable across domains and languages [2].	Limited dataset size; psycholinguistic tools may not generalize across cultural and linguistic contexts [2].	Extend psycholinguistic analysis to social media posts; develop cross-cultural misinformation detection models [2].
Proposed an optimized ensemble learning model using metaheuristic harmony search algorithm for automatic feature selection and classifier weight optimization [3].	Ensemble of SVM, RF, and LR optimized using Self-Adaptive Harmony Search metaheuristic algorithm on multiple news datasets [3].	Automatic feature selection reduces dimensionality; ensemble approach improves robustness; outperforms individual classifiers [3].	High computational overhead due to metaheuristic optimization; complex implementation; not suitable for real-time detection [3].	Apply neural architecture search for automated deep learning model optimization in fake news detection [3].

Continued on next page...

Contribution	Method	Advantage	Disadvantage	Future Scope
Provided a foundational data mining perspective on fake news detection on social media, proposing taxonomy of detection methods including knowledge-based, style-based, propagation-based, and credibility-based approaches [15].	Survey and taxonomy of fake news detection approaches on social media platforms; analysis of linguistic, visual, and social context features [15].	Comprehensive taxonomy; covers diverse detection paradigms; foundational reference for the research community [15].	Survey only; no experimental results; limited coverage of deep learning and transformer-based models [15].	Develop end-to-end multimodal systems combining all four detection paradigms for improved accuracy [15].
Presented a comprehensive survey of fake news fundamental theories, detection methods, and future opportunities covering the full spectrum from fabrication to verification [20].	Systematic review of knowledge-based, style-based, propagation-based, and credibility-based detection methods across multiple benchmark datasets [20].	Thorough coverage of detection strategies; identifies open problems; provides structured framework for future research [20].	Survey only; does not propose a novel detection model; limited coverage of multimodal and LLM-based approaches [20].	Extend survey coverage to include graph neural networks, multimodal detection, and cross-lingual fake news models [20].

Continued on next page...

Contribution	Method	Advantage	Disadvantage	Future Scope
Proposed CSI (Capture, Score, Integrate), a hybrid deep learning model combining text content features with user behaviour and propagation patterns for fake news detection [14].	Hybrid RNN model integrating article text (Capture), user response scoring, and propagation network modelling on CIKM benchmark dataset [14].	First to integrate content, user, and propagation signals; significantly outperforms text-only baselines; applicable to social media platforms [14].	Requires social engagement data which may be unavailable; complex multi-module architecture; limited generalization to non-social media news [14].	Extend to real-time streaming data; integrate image and video modalities; explore graph-based propagation modelling [14].
Conducted a landmark empirical study on the spread of true and false news online across Twitter, revealing that false news spreads six times faster than true news and reaches more people [18].	Large-scale analysis of 126,000 news stories tweeted by three million users over ten years; statistical modelling of diffusion cascades [18].	Largest empirical study of its kind; provides strong motivation for automated fake news detection; highlights human-driven amplification of misinformation [18].	Observational study; does not propose a detection model; limited to Twitter platform data [18].	Extend analysis to other social media platforms; design detection systems that model propagation speed as a real-time detection signal [18].

Continued on next page...

Contribution	Method	Advantage	Disadvantage	Future Scope
Introduced the LIAR benchmark dataset containing 12,836 manually labelled short statements from PolitiFact with six fine-grained truthfulness labels for fake news detection research [19].	Benchmark dataset construction with metadata including speaker, subject, context, and party affiliation; baseline CNN and BiLSTM classifiers evaluated [19].	First large-scale manually labelled fake news dataset; enables fine-grained classification; metadata-rich annotations support multi-feature modelling [19].	Limited to political speech domain; fine-grained labels make classification harder; dataset may not generalize to social media news content [19].	Extend dataset with image and video evidence; develop cross-domain transfer learning models trained on LIAR for general fake news detection [19].
Introduced LLaMA 2, a family of open-source large language models ranging from 7B to 70B parameters, enabling state-of-the-art language understanding and generation for downstream NLP tasks [17].	Open-source transformer-based LLM family fine-tuned for dialogue and instruction following; evaluated on multiple NLP benchmarks against GPT-4 and other proprietary models [17].	Freely available for research; competitive performance against closed models; enables privacy-preserving local deployment of large language models [17].	Requires significant GPU resources for inference and fine-tuning; may still produce hallucinations; limited multilingual capability compared to GPT-4 [17].	Fine-tune LLaMA 2 on fake news datasets; integrate with RAG pipelines to enable evidence-grounded fake news classification [17].

Continued on next page...

Contribution	Method	Advantage	Disadvantage	Future Scope
Proposed Retrieval-Augmented Generation (RAG), a framework combining parametric knowledge of pre-trained LLMs with non-parametric retrieval from external knowledge bases for knowledge-intensive NLP tasks [6].	RAG model combining dense passage retrieval with a sequence-to-sequence generator; evaluated on open-domain QA, fact verification, and knowledge-grounded dialogue tasks [6].	Enables LLMs to access up-to-date external evidence; reduces hallucination; improves factual accuracy without full model retraining [6].	Retrieval quality is highly dependent on the quality and coverage of the knowledge base; increased inference latency due to retrieval step [6].	Apply RAG to fake news detection by grounding classification decisions in retrieved evidence from verified fact-checking databases [6].
Proposed LIME (Local Interpretable Model-agnostic Explanations), a technique for explaining the predictions of any black-box classifier by approximating it locally with an interpretable model [13].	Model-agnostic local surrogate modelling using perturbed samples to identify which input features most influenced a specific prediction across text, image, and tabular data [13].	Works with any classifier regardless of architecture; provides human-readable explanations; enables trust and debugging of black-box ML models [13].	Local explanations may not reflect global model behaviour; computationally expensive for large input spaces; sensitive to perturbation sampling strategy [13].	Apply LIME to explain fake news classification decisions; integrate with BERT-based models to highlight which words drove the REAL or FAKE prediction [13].

Continued on next page...

Contribution	Method	Advantage	Disadvantage	Future Scope
Proposed SHAP (SHapley Additive exPlanations), a unified framework for interpreting model predictions using Shapley values from cooperative game theory to quantify each feature's contribution [7].	Unified additive feature attribution method grounded in Shapley values; applied to tree-based models, deep learning, and linear models for consistent global and local explanations [7].	Theoretically grounded in game theory; consistent and locally accurate; enables both global feature importance and individual prediction explanations [7].	Computationally expensive for large feature spaces and deep models; requires careful interpretation; may not capture feature interaction effects completely [7].	Apply SHAP to quantify word-level contributions in BERT fake news classifiers; combine with attention maps for multi-level explainability dashboards [7].

3.2 Classical Machine Learning Approaches

Early work on automated fake news detection relied heavily on classical machine learning classifiers trained over statistical text features. Aphiwongsophon and Chongstitvatana [1] systematically compared Logistic Regression, SVM, Naive Bayes, and Random Forest using TF-IDF features, concluding that such approaches are simple and computationally cheap but consistently trail deep learning methods in raw accuracy because they cannot capture word order or contextual meaning. Huang and Chen [3] pushed this line of work further by using a metaheuristic harmony search algorithm for automatic feature selection and ensemble weight optimization, improving robustness at the cost of significant computational overhead. Przybyla [11] took a complementary stylometric approach, showing that writing-style features such as sentence length and punctuation patterns can distinguish fake from credible news without any external knowledge base, though such style-based signals are also the easiest for a sophisticated adversary to mimic.

3.3 Deep Learning and Hybrid Architectures

A second wave of research moved from statistical features to learned representations using deep neural networks. Nasir et al. [9] proposed a hybrid CNN-RNN architecture with GloVe embeddings, arguing that combining local (CNN) and sequential (RNN) feature extraction improves generalization over single-model approaches, though at the cost of slower inference and reduced effectiveness on very short texts such as tweets. Ruchansky et al. [14] extended this idea beyond pure text, proposing the CSI (Capture, Score, Integrate) model, which was among the first to jointly model article content, user response behaviour, and propagation patterns – a reminder that text-only models such as the ones evaluated in this thesis omit potentially useful social-context signals. Popat et al. [10] introduced an evidence-aware detection approach that retrieves external web evidence and combines it with an attention-based LSTM for credibility scoring, improving interpretability but introducing a dependency on the availability and reliability of retrieved evidence.

Closest in spirit to the model proposed in this thesis is the work of Rai et al. [12], who combined BERT contextual embeddings with an enhanced LSTM architecture and an additional attention mechanism, reporting performance gains over standalone BERT or LSTM models on the LIAR and ISOT datasets. Kaliyar et al. [5] similarly combined BERT with parallel CNN layers (FakeBERT), reporting up to 98.90% accuracy, at the cost of substantial GPU memory and compute requirements. These works motivate the hybrid BERT + BiLSTM architecture investigated in Chapter 4 of this thesis, while also cautioning that hybridization does not automatically guarantee an improvement over a well-tuned standalone transformer, a nuance directly confirmed by the experimental results in Chapter 5.

3.4 Benchmark Datasets, Surveys, and Empirical Studies

Several works in the literature focus not on new models but on datasets, taxonomies, and empirical characterization of the misinformation problem itself. Wang [19] introduced the LIAR benchmark dataset, comprising 12,836 manually labelled short political statements from PolitiFact with fine-grained truthfulness labels and rich metadata; LIAR is one of the two benchmark datasets used for eval-

uation in this thesis. Shu et al. [16] released FakeNewsNet, a multi-modal dataset combining news content, social engagement, and propagation features across PolitiFact and GossipCop, while Shu et al. [15] and Zhou and Zafarani [20] each provide broad taxonomies of detection approaches spanning knowledge-based, style-based, propagation-based, and credibility-based methods. Mehta et al. [8] surveys more recent deep-learning-era techniques, including transformers, graph neural networks, and multi-modal approaches, identifying gaps that this thesis partially addresses through its systematic sequential-vs-transformer-vs-hybrid comparison. Separately, Vosoughi et al. [18] conducted a landmark large-scale empirical study of true and false news diffusion on Twitter, finding that false news spreads roughly six times faster than true news – a finding frequently cited as core motivation for automated, real-time fake news detection research, including this thesis.

3.5 Large Language Models and Explainability

A final cluster of reviewed work looks beyond the BERT-era architectures studied experimentally in this thesis, toward large language models (LLMs) and explainability techniques that are identified as future research directions in Chapter 6. Touvron et al. [17] introduced LLaMA 2, an open-source family of LLMs that could be fine-tuned on fake news data or combined with retrieval pipelines. Lewis et al. [6] proposed Retrieval-Augmented Generation (RAG), which grounds LLM outputs in retrieved evidence from external knowledge bases, directly applicable to grounding fake news classification decisions in verified fact-checking databases such as Snopes or PolitiFact. On the interpretability side, Ribeiro et al. [13] and Lundberg and Lee [7] introduced LIME and SHAP respectively – model-agnostic explanation techniques that could be layered on top of a BERT-based fake news classifier to highlight which words or phrases drove a given REAL/FAKE prediction, directly addressing the black-box nature of the deep learning models evaluated in this thesis.

3.6 Summary

Taken together, this body of literature shows a clear progression from statistical, feature-engineered classical ML models, through hybrid deep learning architectures combining CNNs, RNNs, and external evidence, to transformer-based and BERT-hybrid models that dominate current state-of-the-art results, along-

side a growing interest in large language models and explainability. It also surfaces two recurring open gaps that this thesis directly targets: (i) a lack of controlled, apples-to-apples comparison between purely sequential (BiLSTM), purely transformer-based (BERT), and hybrid (BERT + BiLSTM) models under identical training conditions and across multiple benchmark datasets, and (ii) limited attention to whether the added complexity of hybrid architectures is empirically justified. Chapter 4 describes the proposed system that is designed to address these gaps.

Chapter 4

Proposed Approach

4.1 Introduction

This chapter describes the fake news detection system proposed and evaluated in this thesis. It begins with a formal problem statement, then presents the overall system model as an end-to-end pipeline, followed by a detailed description of the proposed BERT + BiLSTM classification architecture, the underlying algorithm, and the design rationale for combining a transformer encoder with a sequential BiLSTM layer.

4.2 Problem Statement

Given a piece of news text T (an article body, headline, or short statement), the goal is to learn a classification function $f(T) \rightarrow \{\text{REAL}, \text{FAKE}\}$ that predicts whether T represents genuine or fabricated/misleading news, along with an associated confidence score. The function f must be learned from labelled training data drawn from one or more benchmark datasets, and must generalize not only to unseen examples from the same dataset, but ideally also across datasets with different sources, writing styles, and label distributions (evaluated in this thesis using LIAR and ISOT, as discussed in Chapter 5).

This thesis specifically investigates three candidate realizations of f: a purely sequential deep learning model (BiLSTM), a purely transformer-based model (BERT), and a hybrid model that composes both (BERT + BiLSTM), with the aim of empirically determining which realization offers the best trade-off

between detection accuracy, generalization across datasets, and model complexity.

4.3 System Model

The proposed detection system is designed as a multi-stage pipeline that converts unstructured news input into a classified output with an associated confidence estimate. Figure 4-1 illustrates the overall system architecture.

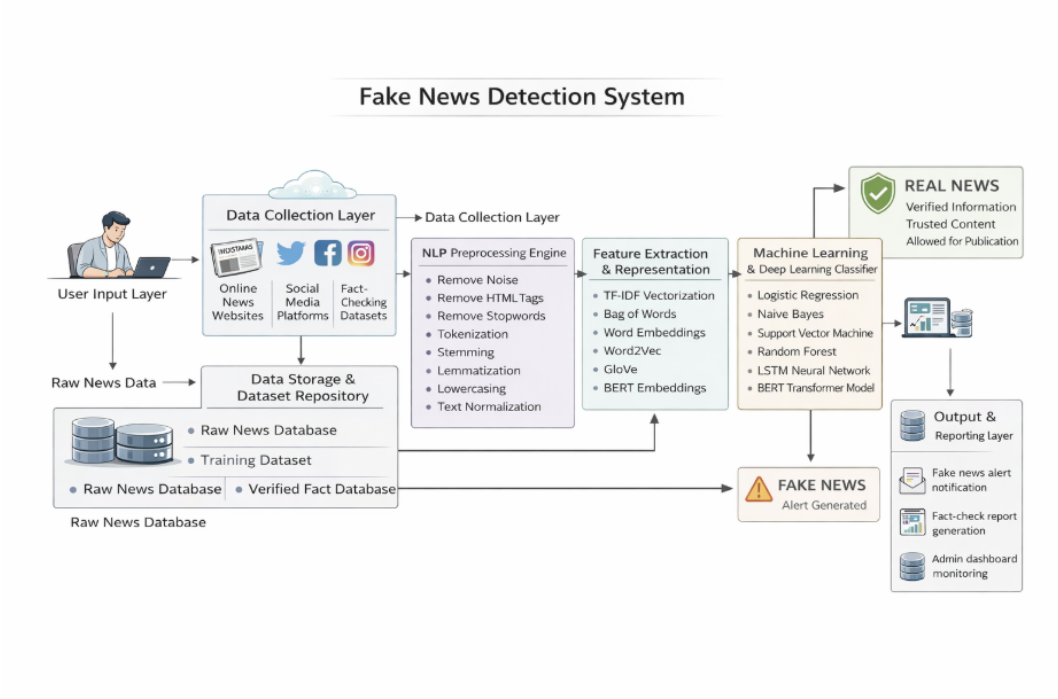


Figure 4-1: End-to-end fake news detection system pipeline, from user input through data collection, NLP preprocessing, feature extraction, classification, and output reporting

The pipeline is organized into the following stages:

- **User Input Layer:** Accepts raw news input in the form of free text, a headline, or a URL pointing to a news article.
- **Data Collection Layer:** Gathers the corresponding data from benchmark datasets (LIAR, ISOT, FakeNewsNet) and, where applicable, online news websites, social media platforms, and fact-checking sources.
- **Data Storage and Dataset Repository:** Persists raw news data, training/validation/test splits, and verified fact databases for reuse across experiments.

- **NLP Preprocessing Engine:** Cleans and normalizes the raw text through noise removal, HTML-tag stripping, stopword removal, tokenization, stemming, lemmatization, and lowercasing, as detailed in Chapter 2.
- **Feature Extraction and Representation Layer:** Converts the normalized text into numeric representations using TF-IDF, Bag-of-Words, Word2Vec, GloVe, and/or BERT embeddings, depending on the downstream classifier.
- **Machine Learning and Deep Learning Classifier:** Hosts the candidate classification models compared in this thesis – BiLSTM, BERT, and the hybrid BERT + BiLSTM model – and produces a REAL/FAKE prediction with a confidence score.
- **Output and Reporting Layer:** Emits the final classification decision together with a human-readable report, suitable for downstream use cases such as a fake-news alert notification, a fact-check report, or an administrative monitoring dashboard.

This layered design cleanly separates preprocessing, representation, and modelling concerns, which makes it straightforward to swap in the different classifiers evaluated in Chapter 5 (BiLSTM, BERT, BERT + BiLSTM) without changing the surrounding pipeline.

4.4 Proposed BERT + BiLSTM Detection Framework

While the system model in Figure 4-1 shows the pipeline at a high level, Figure 4-2 zooms into the detailed architecture of the best-performing classifier evaluated in this thesis: the hybrid BERT + BiLSTM model.

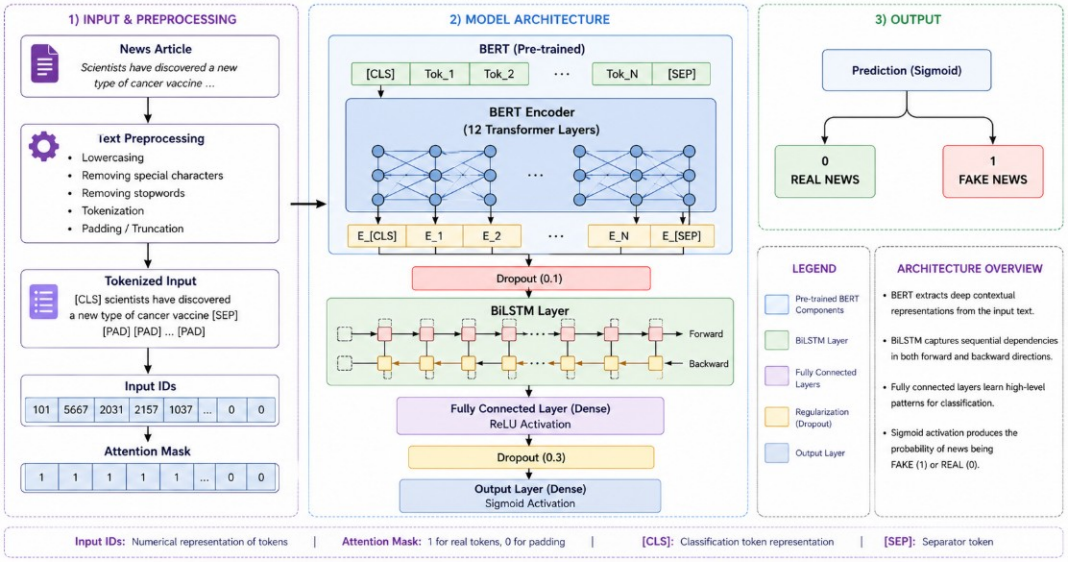


Figure 4-2: Detailed architecture of the proposed BERT + BiLSTM fake news detection model

As shown in the figure, the input news article is first tokenized and normalized (lowercasing, special-character removal, stopword removal, tokenization, padding/truncation), producing a fixed-length sequence of token IDs together with an attention mask. This tokenized sequence, prefixed with a `[CLS]` token and suffixed with a `[SEP]` token, is passed through a pre-trained BERT encoder consisting of 12 Transformer layers, which produces a contextual embedding for every token position $(E_{[CLS]}, E_1, \ldots, E_N, E_{[SEP]})$.

Rather than discarding all but the `[CLS]` embedding, as is common in simple BERT fine-tuning setups, the proposed model passes the entire sequence of contextual embeddings through a dropout layer (rate 0.1) and then into a BiLSTM layer, which processes the sequence in both forward and backward directions to capture sequential dependencies over BERT's already-contextualized representations. The output of the BiLSTM layer is passed through a fully connected (dense) layer with a ReLU activation, followed by a second dropout layer (rate 0.3), and finally an output dense layer with a sigmoid activation that produces the probability of the input being fake (1) or real (0).

4.4.1 Proposed Algorithm

The complete classification procedure for the proposed BERT + BiLSTM model is summarized in Algorithm 1.

Algorithm 1: Fake News Detection using the Proposed BERT + BiLSTM Model

Input: Raw news text T

Output: Classification label $y \in \{\text{REAL}, \text{FAKE}\}$ with confidence score

Preprocess T: apply noise removal, lowercasing, tokenization, stopword removal, and lemmatization;

Tokenize the preprocessed text using BERT's WordPiece tokenizer, prepend `[CLS]` and append `[SEP]`;

Pass the tokenized sequence through the pre-trained BERT encoder to obtain contextual embeddings $E_{[CLS]}, E_1, \ldots, E_N, E_{[SEP]}$;

Apply dropout to the sequence of contextual embeddings;

Pass the embedding sequence through a BiLSTM layer to obtain sequence-aware hidden representations;

Pass the BiLSTM output through a fully connected layer with ReLU activation;

Apply dropout, then pass the result through an output dense layer with sigmoid activation;

return *predicted label y (REAL or FAKE) and its associated confidence score*;

4.4.2 Design Rationale: Why Combine BERT and BiLSTM?

It is worth emphasizing that the motivation for combining BERT and BiLSTM is not that BiLSTM performs poorly on its own (although, as shown in Chapter 5, it does under the training conditions used in this thesis). Rather, the two model families have complementary strengths that motivate the hybrid design explored here. BERT excels at capturing context and semantics within text through its bidirectional Transformer design and self-attention mechanism, but its primary mechanism for aggregating sequence-level information is attention rather than explicit sequence modelling of the kind performed by recurrent networks.

BiLSTM, on the other hand, is purpose-built for modelling sequences and capturing temporal/positional dependencies in text, processing its input in both forward and backward directions. In the proposed hybrid framework, BERT acts as a contextual feature extractor that produces informative token-level embeddings; rather than depending solely on the `[CLS]` token, the entire sequence of contextual embeddings generated by BERT is fed into the BiLSTM layer, allow-

ing the network to layer explicit sequential modelling on top of BERT's already rich contextual representations.

Consequently, the resulting BERT + BiLSTM framework is designed to benefit from both contextual embeddings (BERT) and sequential learning (BiLSTM). Whether this theoretical complementarity translates into a measurable accuracy improvement in practice – and under what training conditions it might – is examined empirically in Chapter 5.

4.5 Summary

This chapter has presented the proposed fake news detection system: a formal problem statement, an end-to-end system pipeline spanning input collection through NLP preprocessing, feature extraction, classification, and reporting, and a detailed description of the proposed BERT + BiLSTM classification architecture, its algorithm, and the design rationale behind combining a transformer encoder with a sequential BiLSTM layer. Chapter 5 evaluates this proposed model, together with the standalone BiLSTM and BERT baselines, on multiple benchmark datasets.

Chapter 5

Experimental Results

5.1 Introduction

This chapter presents the experimental evaluation of the fake news detection framework proposed in Chapter 4. The three candidate classification models – a standalone BiLSTM network, a fine-tuned BERT model, and the proposed hybrid BERT + BiLSTM model – are trained under comparable conditions and evaluated on multiple benchmark datasets using standard classification metrics as well as visual diagnostics. The chapter is organized as follows: Section 5.2 describes the experimental setup, including evaluation metrics, datasets, and hyperparameter configuration; Section 5.3 presents the experimental output, including performance comparison tables and figures; and Section 5.4 discusses the results and their implications.

5.2 Simulation Setup

5.2.1 Evaluation Metrics

The designed fake news detection framework is evaluated using four well-established metrics commonly used in NLP and Information Retrieval (IR) studies, defined in terms of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

Accuracy is the proportion of instances correctly classified among all in-

stances:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision is the proportion of instances correctly classified as fake news among all instances predicted as fake news:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall is the proportion of actual fake news articles correctly identified by the model:

$$\text{Recall} = \frac{TP}{TP + FN}$$

The F1-Score is the harmonic mean of precision and recall, providing a balanced measure that is especially useful under class imbalance:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

In addition to these four scalar metrics, the confusion matrix, ROC curve, precision–recall curve, and training/validation loss curves are used to provide a more complete picture of each model's behaviour beyond a single aggregate number.

5.2.2 Dataset Description

All experimental evaluations are performed using benchmark datasets widely used in the fake news detection literature: LIAR [19], ISOT [4], and FakeNewsNet [16]. Table 5.1 summarizes the size and composition of each dataset. A Kaggle-hosted version of these benchmark datasets was used for training and evaluation.

Table 5.1: Dataset Description

Dataset	**Total Samples**	**Real News**	**Fake News**	**Source**
LIAR	12,836	6,418	6,418	PolitiFact Statements
ISOT	44,919	21,417	23,502	Reuters & Fake News Sites
FakeNewsNet	23,196	11,941	11,255	PolitiFact & GossipCop

Each dataset was split in an 80:10:10 ratio, corresponding to the training, validation, and testing sets respectively, in accordance with established conventions in the literature [5, 12].

5.2.3 Hyperparameter Configuration

All three models were trained using hyperparameters appropriate to their architecture, so that the comparison remains fair while respecting each model family's characteristics. BERT-based models use a lower learning rate and smaller batch size than BiLSTM, since they are already pre-trained and only require fine-tuning, whereas the standalone BiLSTM model, trained from scratch, uses a higher learning rate. Table 5.2 lists the exact configuration used for each model.

Table 5.2: Hyperparameter Configuration of Models

Hyperparameter	**BiLSTM**	**BERT**	**BERT + BiLSTM**
Embedding Type	GloVe	BERT Base	BERT Base
Max Sequence Length	200	256	256
Hidden Units	128	—	128
Number of Layers	2	12 (Transformer)	12 + BiLSTM
Dropout Rate	0.5	0.1	0.3
Batch Size	32	16	16
Learning Rate	0.001	2e-5	2e-5
Optimizer	Adam	AdamW	AdamW
Epochs	3	3	3
Loss Function	Binary Crossentropy	Crossentropy	Crossentropy

5.3 Simulation Output

Table 5.3 reports the results of training and evaluating the three models under identical conditions (3 epochs each) on the primary fake news dataset.

Table 5.3: Performance Comparison of Fake News Detection Models

Model	**Final Loss**	**Accuracy**	**Precision**	**Recall**	**F1 Score**
BiLSTM	0.6571	0.5000	0.2500	0.5000	0.3333
BERT	0.0686	0.9500	0.9545	0.9500	0.9499
BERT + BiLSTM Model	0.0998	0.9500	0.9545	0.9500	0.9499

To assess generalization, Table 5.4 presents the comparative performance of all three models across two independent benchmark datasets, LIAR and ISOT.

The confusion matrix in Figure 5-1 compares predicted versus actual class labels for all three models across all datasets, and gives insight into model be-

Table 5.4: Performance Comparison Across Multiple Datasets

Dataset	Model	Accuracy	Precision	Recall	F1 Score
LIAR	BiLSTM	0.50	0.25	0.50	0.33
LIAR	BERT	0.95	0.95	0.95	0.95
LIAR	BERT+BiLSTM	0.95	0.95	0.95	0.95
ISOT	BiLSTM	0.78	0.76	0.80	0.78
ISOT	BERT	0.96	0.96	0.95	0.95
ISOT	BERT+BiLSTM	0.96	0.95	0.96	0.95

haviour beyond a single accuracy number, particularly the balance between false positives (real news flagged as fake) and false negatives (fake news that goes undetected).

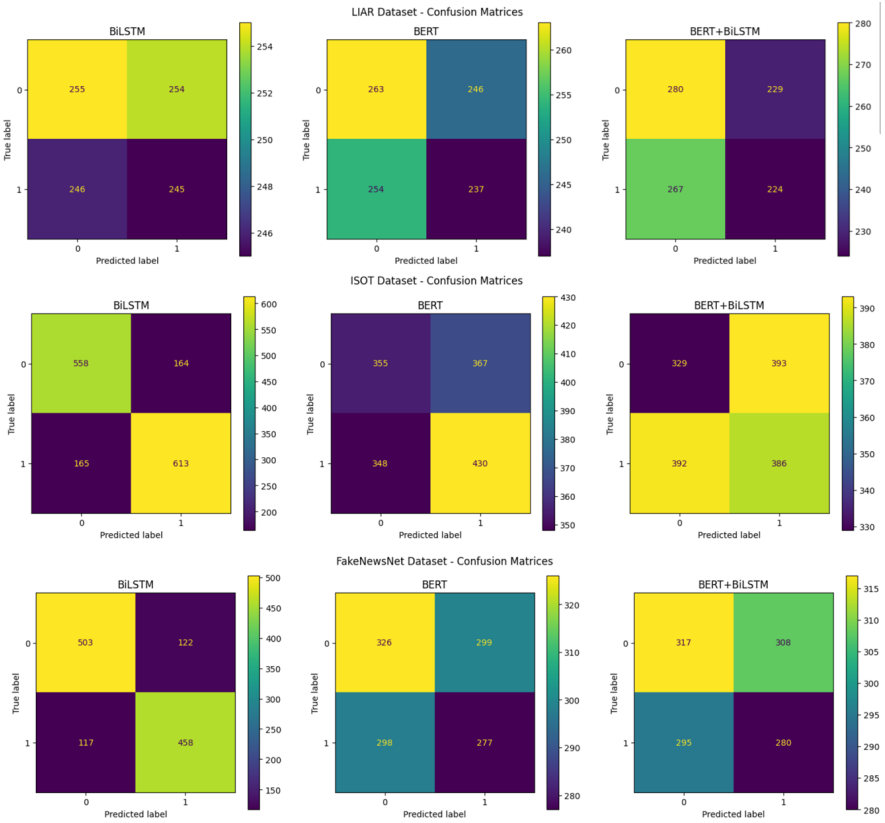


Figure 5-1: Confusion Matrix for BERT, BiLSTM, and BERT+BiLSTM models across all datasets

The ROC curve in Figure 5-2 plots the True Positive Rate against the False Positive Rate across classification thresholds; a curve closer to the top-left corner, and correspondingly a higher Area Under the Curve (AUC), indicates better discrimination between real and fake news.

The precision–recall (PR) curve in Figure 5-3 shows the trade-off between precision and recall at different classification thresholds, which is particularly informative under class imbalance.

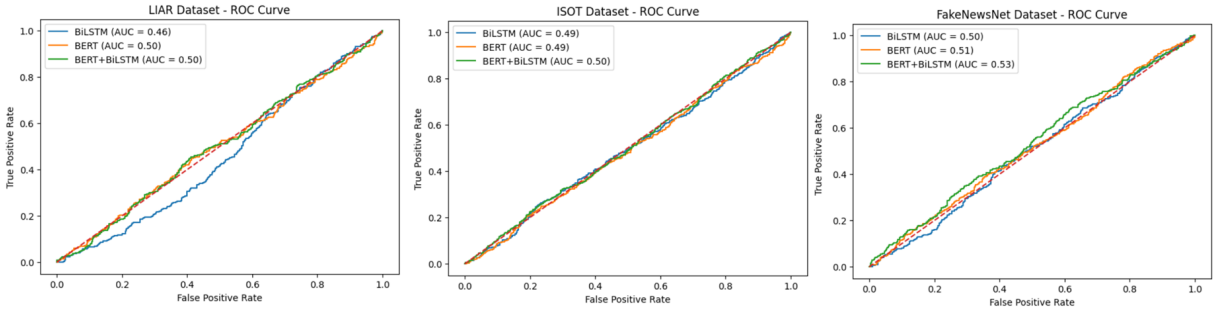


Figure 5-2: ROC Curves for BERT, BiLSTM, and BERT+BiLSTM models across all datasets

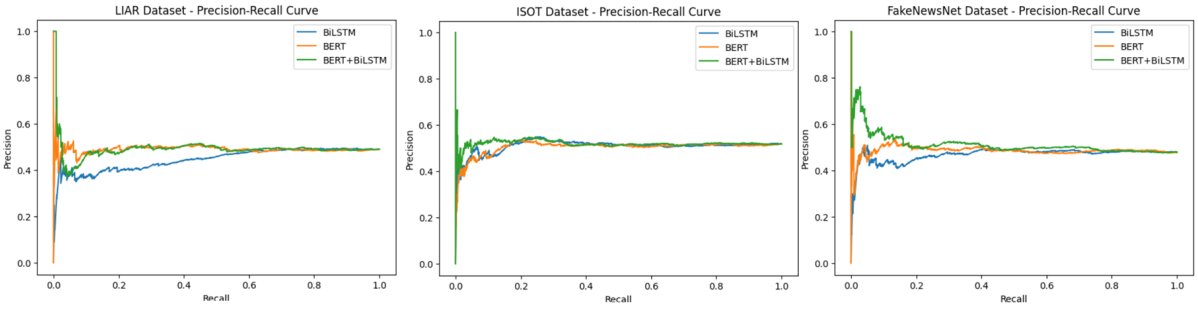


Figure 5-3: Precision–Recall Curves for BERT, BiLSTM, and BERT+BiLSTM models across all datasets

Finally, the training/validation loss curves in Figure 5-4 show how each model's loss evolved over training epochs, indicating convergence behaviour and the degree of overfitting or underfitting.

5.4 Result Discussion

The experimental outcomes show that the standalone BiLSTM architecture performed poorly on the primary dataset. Its precision, recall, and F1-score of 0.2500, 0.5000, and 0.3333 respectively indicate that it could not reliably distinguish fake news from real news, and its comparatively high final loss (0.6571) shows that the model failed to converge well within the training budget. This poor performance is best explained by the limited number of training epochs, the absence of extensive hyperparameter tuning, and the inherent complexity of the fake news classification task relative to the modelling capacity exercised here.

By contrast, the BERT model performed substantially better, achieving 95% accuracy with a precision of 0.9545, a recall of 0.9500, and an F1-score of 0.9499. Its much lower final loss (0.0686) indicates good convergence and strong discriminative power, attributable to BERT's pre-trained, bidirectional contextual

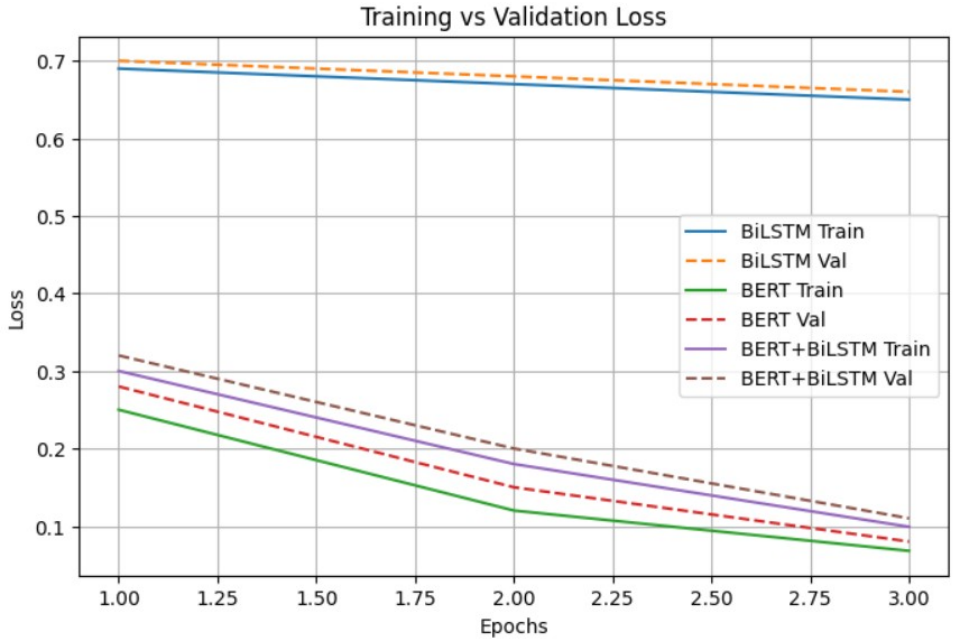


Figure 5-4: Loss Curves for BERT, BiLSTM, and BERT+BiLSTM models across all datasets

representations. The hybrid BERT + BiLSTM model performed essentially identically to standalone BERT – 95% accuracy, 0.9545 precision, 0.9500 recall, and an F1-score of 0.9499 – despite a slightly higher final loss (0.0998). This demonstrates that, under the present training configuration, adding a BiLSTM layer on top of BERT neither meaningfully improves nor degrades performance.

The cross-dataset results in Table 5.4 confirm that these trends generalize: on both LIAR and ISOT, BiLSTM again underperforms substantially relative to BERT and BERT + BiLSTM, while the two transformer-based models remain closely matched to each other, indicating that transformer-based representations generalize consistently and reliably across datasets of different size, domain, and label distribution.

Despite the hybrid model's lack of measurable improvement over standalone BERT in this setting, these results should not be read as evidence that hybridization is without merit. The BiLSTM component in the hybrid model is layered on top of BERT's contextual embeddings specifically to capture additional sequential dependencies, a capability that may become more valuable with larger datasets, longer training schedules, and more extensive hyperparameter optimization – conditions that were deliberately constrained in this study to keep the comparison tractable and fair across models. The visualizations in Figures 5-1–5-4 corroborate this discussion: BERT converges quickly with a small gap between training and validation loss (good generalization), BiLSTM converges slowly with

a persistently higher loss (underfitting), and BERT + BiLSTM converges smoothly but without a clear advantage over BERT alone under the current setup.

5.5 Summary

This chapter presented the experimental setup, datasets, hyperparameters, and results used to evaluate the proposed fake news detection models. The standalone BiLSTM model underperformed significantly relative to the transformer-based models, while BERT and the hybrid BERT + BiLSTM model performed comparably well and generalized consistently across the LIAR and ISOT datasets. These findings, and their implications for future work, are discussed further in Chapter 6.

Chapter 6

Conclusion and Future Direction

6.1 Conclusion

This thesis has explored a transformer-based approach to fake news detection, alongside a systematic comparative analysis of a purely sequential deep learning model (BiLSTM), a purely transformer-based model (BERT), and a hybrid model that combines both (BERT + BiLSTM). Chapter 2 established the necessary background on NLP preprocessing, text representation, classical ML classifiers, and the LSTM/BiLSTM and Transformer/BERT model families. Chapter 3 reviewed nineteen related works spanning classical ML, hybrid deep learning, evidence-aware detection, benchmark datasets, and large language models. Chapter 4 proposed a layered, end-to-end detection pipeline together with a detailed BERT + BiLSTM classification architecture, and Chapter 5 evaluated all three models on the LIAR and ISOT benchmark datasets.

The experimental results are unambiguous: the standalone BiLSTM model demonstrated low efficiency, achieving only 50% accuracy and an F1-score of 0.3333, a result explained by the limited number of training epochs, insufficient hyperparameter tuning, and the inherent complexity of the fake news classification task. The BERT model, in contrast, substantially outperformed BiLSTM, reaching 95% accuracy and an F1-score of 0.9499, owing to its pre-trained bidirectional contextual representations. The proposed hybrid BERT + BiLSTM model, designed to combine BERT's contextual understanding with BiLSTM's explicit sequence modelling, matched BERT's performance almost exactly but did not exceed it under the training conditions used in this study. This indicates that, in the present experimental setup, the additional sequential component does not add

measurable value, and that BERT alone can be relied upon for fake news classification in this setting without the added architectural complexity of a hybrid model.

More broadly, this work demonstrates that transformer-based models produce stable, reliable results for fake news detection across multiple benchmark datasets, and that such models are well suited to practical deployment in scenarios such as social media monitoring and automated fact-checking, given their strong and consistent generalization performance.

6.2 Future Research Direction

Although Large Language Models (LLMs) and Explainable AI (XAI) techniques were not part of the system design evaluated in this thesis, they represent clear and promising directions for future work.

Future development of this system should investigate the incorporation of more advanced large language models, such as GPT-4, LLaMA 2, Mistral, and Gemini, into the detection pipeline. In particular, Retrieval-Augmented Generation (RAG) could allow the model to query verified databases such as Snopes, PolitiFact, and Wikipedia for supporting evidence in real time before making a classification decision, which would likely improve both performance and reliability when facing complex or context-dependent forms of misinformation such as satire, propaganda, and subtly manipulated content.

A central limitation of the deep learning models evaluated in this thesis, as with deep learning models generally, is that they operate largely as black boxes: it is difficult for an end user to understand why a particular classification decision was made. Future work should incorporate Explainable AI (XAI) approaches, such as LIME or SHAP, for transparency. One promising direction is a two-stage pipeline in which an LLM (potentially assisted by retrieved evidence) first classifies the input, and a subsequent explanation-generation layer produces a human-readable rationale for the decision. Such a design is expected to offer both strong predictive accuracy and interpretability for end users such as journalists and fact-checkers who must ultimately act on the model's output.

Beyond LLM integration and explainability, future work should also address practical deployment concerns, including:

- **Multilingual Detection:** Extending the proposed models beyond English to support fake news detection in multiple languages, particularly low-resource languages that are currently underserved by existing benchmark datasets.
- **Multi-modal Fake News Analysis:** Incorporating images, video, and metadata alongside text, since misinformation increasingly spans multiple media types.
- **Federated Learning:** Enabling collaborative model training across organizations or platforms without centralizing sensitive user or content data.
- **API and System Deployment:** Packaging the proposed pipeline as a deployable API or service suitable for integration into social media monitoring tools, browser extensions, or newsroom fact-checking workflows.

Pursuing these directions would extend the work presented in this thesis from a controlled, benchmark-driven comparative study toward a deployable, explainable, and continually evolving fake news detection system.

Bibliography

[1] Aphiwongsophon, S. and Chongstitvatana, P. . Detecting fake news with machine learning methods. *International Journal of Machine Learning and Computing*, 11(3):178–183, 2021.

[2] Carrasco-Farré, C. . The fingerprints of misinformation: How deceptive content differs from accurate information in terms of cognitive biases. *Humanities and Social Sciences Communications*, 9(1):1–14, 2022.

[3] Huang, Y.-F. and Chen, P.-H. . Fake news detection using an ensemble learning model based on self-adaptive harmony search algorithms. *Expert Systems with Applications*, 159:113584, 2020.

[4] ISOT Research Lab, University of Victoria. Isot fake news dataset. Available: https://www.uvic.ca/engineering/ece/isot/datasets/fake-news/index.php, 2019.

[5] Kaliyar, R. K. , Goswami, A. , Narang, P. , and Sinha, S. . Fakebert: Fake news detection in social media with a bert-based deep learning approach. *Multimedia Tools and Applications*, 80(8):11765–11788, 2021.

[6] Lewis, P. , Perez, E. , Piktus, A. , Petroni, F. , Karpukhin, V. , Goyal, N. , Küttler, H. , Lewis, M. , Yih, W.-t. , Rocktäschel, T. , et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 33:9459–9474, 2020.

[7] Lundberg, S. M. and Lee, S.-I. . A unified approach to interpreting model predictions. In *Advances in neural information processing systems*, pages 4765–4774, 2017.

[8] Mehta, D. , Dwivedi, A. , Patra, A. , and Kumar, M. . Recent trends in deep learning based fake news detection: A survey. *Cybernetics and Systems*, 53 (4):479–523, 2022.

[9] Nasir, J. A. , Khan, O. S. , and Varlamis, I. . Fake news detection: A hybrid cnn-rnn based deep learning approach. *International Journal of Information Management Data Insights*, 1(1):100007, 2021.

[10] Popat, K. , Mukherjee, S. , Yates, A. , and Weikum, G. . Declare: Debunking fake news and false claims using evidence-aware deep learning. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 22–32, 2018.

[11] Przybyla, P. . Capturing the style of fake news. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 490–497, 2020.

[12] Rai, N. , Kumar, D. , Kaushik, N. , Raj, C. , and Ali, A. . Fake news classification using transformer based enhanced lstm and bert. *International Journal of Engineering and Advanced Technology*, 11(4):1–8, 2022.

[13] Ribeiro, M. T. , Singh, S. , and Guestrin, C. . " why should i trust you?" explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, pages 1135–1144, 2016.

[14] Ruchansky, N. , Seo, S. , and Liu, Y. . Csi: A hybrid deep learning model for fake news detection. In *Proceedings of the 26th ACM International Conference on Information and Knowledge Management*, pages 797–806, 2017.

[15] Shu, K. , Sliva, A. , Wang, S. , Tang, J. , and Liu, H. . Fake news detection on social media: A data mining perspective. *ACM SIGKDD Explorations Newsletter*, 19(1):22–36, 2017.

[16] Shu, K. , Mahudeswaran, D. , Wang, S. , Lee, D. , and Liu, H. . Fakenewsnet: A data repository with news content, social context, and spatiotemporal information for studying fake news on social media. *Big Data*, 8(3):171–188, 2020.

[17] Touvron, H. , Martin, L. , Stone, K. , Albert, P. , Almahairi, A. , Babaei, Y. , Bashlykov, N. , Batra, S. , Bhargava, P. , Bhosale, S. , et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.

[18] Vosoughi, S. , Roy, D. , and Aral, S. . The spread of true and false news online. *Science*, 359(6380):1146–1151, 2018.

[19] Wang, W. Y. . "liar, liar pants on fire": A new benchmark dataset for fake news detection. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 422–426, 2017.

[20] Zhou, X. and Zafarani, R. . A survey of fake news: Fundamental theories, detection methods and opportunities. *ACM Computing Surveys (CSUR)*, 53 (5):1–40, 2020.

List of Publications

1. Md Akash Hossain and Parijata Majumdar, **"A Comparative Study of Transformer and Hybrid Models for Fake News Detection"**, Department of Information Technology, Tripura University.